

David Gonzalez-Velez

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EDUCATION

The University of Chicago

Sept. 2017 – Jun. 2021

Bachelor of Science

Molecular Engineering – Quantum Track

The University of Chicago - Graham School

Jun. 2025

Quantum Science, Networking, and Communications Certification

EXPERIENCE

Software Developer - Snap On Credit

May 2025 - Present

- Modernized a legacy PICK Basic collection system used by **300 internal users**, updating interfaces with Python/HTML/CSS and enhancing developer productivity through improved customization and UI consistency
- Designed and deployed a new SMS delivery system, replacing discontinued carrier email-to-text services and enabling **1,000–2,000 automated messages/month** through a dedicated vendor API
- Migrated the full legacy platform to a modern terminal emulator and ported programs into an API-driven architecture, improving system maintainability and enabling the creation of new web-based interfaces

Research Assistant - Loyola University

Oct. 2023 - Present

- Developed and maintained a full-stack facial recognition application utilizing React, Flask, and Vercel; spearheaded the implementation of the CI pipeline using GitHub Actions
- Led the transition to Chakra UI to optimize design, enhance user experience, and **improve system accuracy by 50%**, analyzing facial embedding thresholds for better comparison precision.
- Managed project by organizing stand-ups, establishing a Kanban board for task tracking, and collaborating with team members and stakeholders to ensure progress and alignment with research objectives.

Senior Software Engineer - Smyl Fitness Rx

Jan. 2023 - Sep. 2024

- Spearheaded end-to-end development of functional health screening software, overseeing 2 developer teams and successfully launching the product within 3 months
- Integrated MoveNet pose estimation algorithms to accurately and efficiently assess functional health data, achieving a **50% improvement in screening precision** over the initial version

Software Engineer - Capgemini Financial Services

Aug. 2021 - Oct. 2022

- Developed and maintained data pipelines for a credit card management system
- Managed SQL relational databases on Azure Cloud to support a credit card user application

RELEVANT PROJECTS

AI Math Problem Generator

- Developed a REST API for a math question generator using OpenAI's API and Flask to handle requests given problem category and difficulty
- Leveraged React to create a front end to serve the questions in a user-friendly format, using TeX to render the mathematical equations
- Deployed the application using Heroku and utilized Mezmo for monitoring server-side logs

Game Hub Application

- Leveraged the RAWG public API to implement a game hub application with React
- Deployed the application on Vercel and implemented a CI/CD pipeline to maintain up-to-date functionality

PROGRAMMING LANGUAGES, FRAMEWORKS, AND CERTIFICATIONS

Advanced: Python, R, SQL, C, JavaScript, TypeScript, Qiskit, BASIC, LaTeX, React, Flask, Django, Supabase

Intermediate: Flutter, Terraform

Certifications: Microsoft Azure Administrator Associate (2021)

SKILLS

Technical: PowerPoint, Word, Publisher, Excel

Cloud Computing: AWS, GCP, Azure

Language: Fluent in Spanish at a professional, academic, and conversational level